

Airframe Design - Optimization: Part I

Research Skills: Version Control

As part of this assignment, you will be expected to develop introductory skills in version control. In order to do so, complete the following:

- a. [Register for a free Github account.](#)
- b. [Complete this basic git tutorial.](#)
- c. Add, Commit, and Push your assignment submission to your github repository as part of your submission for this assignment.

Technical Tasks: Preliminary Optimization

- a. Consider the elliptic lift distribution from your last assignment. Create an objective function that seeks this lift distribution using the chord distribution as design variables. You will need to write out your optimization problem in mathematical form.
- b. Implement this objective function. Sweep the design variables to verify the correct behavior of the objective function. Provide plots of the objective function as a function of the design variables.
- c. Calculate the jacobian of your objective function. What is this jacobian telling you about your objective function? How does the optimizer use this information? What is the condition number of the jacobian? What does this tell you about the jacobian?

Writing Tasks: Introduction and Methodology

As you begin these design optimization assignments, you will need to make sure you have clear goals and plans. One of the best ways to organize your thoughts is to write a first draft of the introduction to your final technical report, as well as outline your proposed methodology. Therefore, your main written deliverables (using the provided template) for this assignment are:

- a. Write a first draft of the introduction section of your final report.
- b. Outline your methodology section including the following specific items:
 - Both a word and mathematical description of your optimization problem.
 - Descriptions and ranges for your design variables.
 - Descriptions and ranges for your design constraints.
 - A table including the parameters you plan to hold constant, along with values and units for those parameters.
- c. Draft plots and brief descriptions of your preliminary results.